SS-C-197e
July 16, 1964

SUPERSEDING
Int. Fed. Spec. SS-C-197d (COM-NBS)
June 12, 1963 and
Fed. Spec. SS-C-197b
September 26, 1960

FEDERAL SPECIFICATION

CEMENT, PORTLAND, BLAST-FURNACE SLAG

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two types of hydraulic cement, each with two optional provisions.

1.2 Classification. Portland blast-furnace slag cement shall be of the following types and optional suffixes, as specified (6.2).

1.2.1 *Types.*

Type IS—Portland blast-furnace slag cement.

Type IS-A—Air entraining portland blast-furnace slag cement.

1.2.2 *Suffix.* When moderate sulfate resistance or heat of hydration are required (6.2), the following suffixes shall be added to the type designation:

(MS)—Moderate sulfate resistance.

(MH)—Moderate heat of hydration.

(MS) (MH)—Both moderate sulfate resistance and moderate heat of hydration.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

Federal Specifications:

SS-C-192—Cement, Portland.
UU-S-48—Sacks, Paper, Shipping.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 158—Cements, Hydraulic; Sampling, Inspection and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

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Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Portland blast-furnace slag cement is the product obtained by intimately intergrinding a mixture of portland-cement clinker and granulated blast-furnace slag. The amount of granulated blast furnace slag used shall be such that the slag constituent makes up between 25 and 65 percent by weight of the portland blast furnace slag cement.

3.1.1 *Portland-cement clinker.* The manufacturer may use any type of portland cement clinker suitable for use in manufacture of cements complying with SS-C-192, which in combination with the granulated blast-furnace slag will produce a product meeting the chemical and physical requirements as specified in 3.2 and 3.3, respectively.

3.1.2 *Granulated blast-furnace slag.* Granulated blast-furnace slag is the nonmetallic product consisting essentially of silicates and aluminosilicates of calcium which is developed simultaneously with iron in a blast furnace and is produced by rapidly chilling or quenching the molten material in water, steam, or air.

3.2 Chemical requirements. Portland blast-furnace slag cement of the type specified shall conform to the chemical requirements prescribed in table I. When moderate sulfate resisting (MS) cement is specified (6.2), the tricalcium aluminate ($3CaO.Al_2O_3$) content of the portland-cement clinker shall not exceed 8.0 percent.

TABLE I—*Chemical requirements*

	Maximum, percent
Sulfur trioxide (SO_3)	2.5
Sulfide sulfur (S)	2.0
Insoluble residue	1.0
Loss on ignition	3.0

3.3 Physical requirements. Portland blast-furnace slag cement of type specified shall conform to the physical requirements prescribed in table II. When the type of cement is not specified, the requirements of type IS shall govern.

3.4 Processing additions. Only the following materials in addition to granulated slag shall be added during grinding of the clinker: (1) Water and/or untreated calcium sulfate in amounts such that the limits shown in table I for loss on ignition and sulfur trioxide are not exceeded, and (2) At the option of the manufacturer, nondeleterious materials other than water and/or calcium sulfate in amounts less than one percent. When specified (6.2), the manufacturer shall furnish to the procuring agency a written statement certifying that less than one percent of nondeleterious material, other than water and/or calcium sulfate, and no deleterious material, has been added.

3.5 Air-entraining addition. When air-entraining cement is specified, there shall be interground with the clinker an addition of such a nature and in such an amount that the resulting product complies with the applicable requirements of tables I and II. When specified (6.2), the manufacturer shall furnish a written statement giving the source, amount, and brand name of the addition used. When specified (6.2), the manufacturer shall also furnish test data acceptable to the purchaser showing that an air-entraining portland cement (1) produced with this air-entraining addition, (2) meeting the applicable requirements of tables I

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TABLE II —*Physical requirements*

Type	IS	IS-A	IS(MS)	IS-A(MS)
Fineness:				
Residue, No. 325 sieve, max., percent	12.0	12.0	12.0	12.0
Air permeability: [1, 2]				
Average value, min., sq. cm/g.	2800	2800	2800	2800
Min. value, any one sample in average, sq. cm/g.	2600	2600	2600	2600
Autoclave expansion, max., percent	0.20	0.20	0.20	0.20
Time of setting, Gillmore method:				
Initial, min., minutes	60	60	60	60
Final, max., hours	10	10	10	10
Air content of mortar [2], percent by volume	0—12.0	19.0 ± 3.0	0—12.0	19.0 ± 3.0
Compressive strength, min., p.s.i.[3]:				
3 days	1200	900	1000	750
7 days	2100	1500	1800	1400
28 days	3500	2800	3500	2800
Heat of hydration[4]:				
7 days, max., cal/g.	70	70	70	70
28 days, max., cal/g.	80	80	80	80
False set[1], early stiffening of cement paste, ratio of final to initial penetration, percent min.	50	50	50	50

[1] These limits apply only when specified (see 6.2).

[2] Portland blast-furnace slag cement has a lower apparent specific gravity than portland cement, and therefore when the air permeability apparatus is used, the weight of the sample shall be adjusted so that a firm, hard bed is formed by the compacting process, and the fineness shall be calculated by the appropriate formula, using the determined specific gravity. Air content of the mortar shall be calculated by the general formula using the determined specific gravity of the cement if it differs from 3.15 by more than 0.05.

[3] Unless otherwise specified (see 6.2), compressive strength tests will be made only at 3 and 7 days. The strength at any age shall be greater than the strength at the next preceding age.

[4] Applicable only when moderate heat of hydration (MH) cement is specified, in which case the strength requirements shall be 80 percent of the values shown in the table.

and II of SS-C-192, and (3) manufactured not more than three years prior to the date of request, had a durability factor of not less than 80 when tested in accordance with the procedures of method 4201 of Fed. Test Method Std. No. 158.

3.6 Source, amount, and composition. When specified (6.2), the manufacturer shall furnish to the procuring agency a written statement giving the source, amount, and composition of the granulated blast-furnace slag used in the finished cement, and the composition of the portland cement clinker constituent. The composition of the finished cement shall conform to that shown in the statement within the following tolerances: SiO_2 ± 3 percent, Al_2O_3 ± 2 percent, CaO ± 3 percent.

3.7 Identification marking. The brand, type, and net weight shall be clearly marked on each bag.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Inspection responsibility. Cement for use on Federal Government projects will be sampled at the mill, and will be tested for conformance to the specified requirements by the National Bureau of Standards, or other designated Government agency. Sampling and testing by a Government agency may be waived at the discretion of the purchasing agency, and the cement accepted upon the manufacturer's certification of compliance with specification requirements accompanied by a mill test report.

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4.2 Sampling.

4.2.1 *Lot.* Unless otherwise specified (6.2), a lot shall consist of each type of cement submitted for inspection at the same time.

4.2.2 *Sampling for examination of packing.* Packed cement shall be sampled in accordance with level S-3 of MIL-STD-105.

4.2.3 *Sampling for test.* Portland blast-furnace slag cement shall be sampled in accordance with Section 4 of Fed. Test Method Std. No. 158.

4.3 Inspection and tests.

4.3.1 *Inspection.* Packed cement selected in 4.2.2 shall be examined in accordance with table III. The acceptable quality levels (AQL's) for defects listed in table III shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects in accordance with MIL-STD-105.

4.3.2 *Lot acceptance tests.* Samples selected in 4.2.3 shall be subjected to the tests in 4.4.

TABLE III.—*Classification of defects*

Defects	Major	Minor
Bags not of the grade specified	X	
Bags not fully closed	X	
Bags split or torn	X	
Hole in bags	X	
Marking omitted or illegible	X	
Marking incomplete		X
Marking in wrong location		X

4.4 Tests.

4.4.1 *Chemical.* Chemical tests for the requirements in table I shall be conducted in accordance with the applicable test method of Standard No. 158. The tricalcium aluminate ($3CaO.Al_2O_3$) content of the portland cement clinker shall be calculated, when required, from the chemical analysis of the clinker, using the values of Fe_2O_3 and Al_2O_3 determined to the nearest 0.01 percent, and reported to the nearest 0.1 percent, as follows: Tricalcium aluminate = (2.650 x percent Al_2O_3)—(1.692 x percent Fe_2O_3).

4.4.2 *Physical.* Physical tests for the requirements in table II shall be conducted in accordance with the applicable test method of Fed. Test Method Std. No. 158.

4.5 Rejection and retests. The cement shall be rejected if it fails to meet any of the requirements of this specification. Cement remaining in storage prior to shipment for a period longer than 6 months after test may be retested and shall be rejected if it fails to meet any requirements of this specification. Cement having an air content between 12 and 16 percent as determined by the air-entrainment test may be accepted at the discretion of the purchaser when air-entraining concrete is to be used and the air content of the concrete is controlled at the mixer. Cement failing to conform to the limit on autoclave expansion test may be accepted if it passes in a retest of new samples taken within 28 days of the time of the original sampling. The provisional acceptance of the cement at the mill shall not deprive the purchaser of the right of rejection on a retest of autoclave expansion at the time of the delivery of the cement to the purchaser.

4.6 Inspection of preparation for delivery. The packing and marking of the cement shall be examined to determine compliance with the requirements of section 5.

5. PREPARATION FOR DELIVERY

(For civil agency procurement, the definitions and applications of the levels of packing shall be in accordance with Fed. Std. No. 102.)

5.1 Packing. Packing shall be level A, B, or C, as specified (6.2).

5.1.1 *Level A.* Cement in quantities of 94 pounds net shall be packed in bags conforming to UU-S-48, X grades.

5.1.2 *Level B.* Cement in quantities of 94 pounds net shall be packed in bags conforming to UU-S-48, L/W grade.

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5.1.3 *Level C.* Cement shall be packed to insure carrier acceptance and safe delivery to destination in containers complying with the rules and regulations applicable to the mode of transportation.

5.2 **Marking.**

5.2.1 *Civil agencies.* In addition to any special marking specified in the contract or order, the shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 *Military activities.* In addition to any special markings specified in the contract or order, the shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 **Intended use.** Both types of cement are intended for use in general concrete construction.

6.2 **Ordering data.** Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type of cement required (1.2), when not specified, type IS should be furnished.
(*c*) When required, fineness, air permeability (table II, note 1).
(*d*) When required, moderate sulfate resistance (1.2.2, and 3.2).
(*e*) When required, moderate heat of hydration (1.2.2, table II, note 4).
(*f*) When required, limit on false set (table II, note 1).
(*g*) When required, compressive strength at later ages (table II, note 3).
(*h*) When required, a written statement on nondeleterious material (3.4).
(*i*) When required, a written statement on the source, amount, and brand name of addition (3.5).
(*j*) When required, test data on air-entraining addition (3.5).
(*k*) When required, a written statement on the source, amount, and composition of the granulated blast-furnace slag (3.6).
(*l*) Size of lot, if different from 4.2.1.
(*m*) Level of packing required (5.1).

6.3 A 94 pound bag of portland blast-furnace slag cement has a larger volume than a 94 pound bag of portland cement, which may be taken into consideration in purchasing cement, and in proportioning concrete mixtures.

6.4 **Transportation description.** The transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Cement, Portland:
Carload minimum weight 50,000 pounds.

Motor:

Cement, Portland:
Truckload minimum weight 50,000 pounds, subject to Rule 115, National Motor Freight Classification.

6.5 This material is similar to the cement described in ASTM C205-62T.

MILITARY INTERESTS:

Army—MO

Navy—Docks

Air Force—MOA

Copies of this specification may be purchased for cents each.

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